Point Defects in Crystals: Guided Exercises from Freysoldt *et al.* (2014)

Duy M. Dao

November 10, 2026

Purpose and style of this sheet. Exercises 1–13 turn the equations of C. Freysoldt *et al.*, "First-principles calculations for point defects in solids", Rev. Mod. Phys. **86**, 253 (2014), cited as [F], into guided derivations. Each part is a question that leads to the next step, and the exercise ends at a numbered equation of [F]. Most questions use only Sections 1–5, 10 and 11 of the note. Exercises 5, 6 and 9 introduce the Fermi–Dirac and Bose–Einstein distributions, and Exercise 12 introduces diffusion. Both topics lie beyond the note, so these exercises build the needed results from the canonical and grand canonical ensembles.

Units and notation. The Boltzmann constant k_B is kept explicit, and $\beta = 1/(k_B T)$. Where a problem needs the classical ideal gas, the canonical partition function of N identical particles of mass m in the volume V is given as $Z_N = V^N/(N!\,\lambda_{\mathrm{th}}^{3N})$, with the thermal wavelength $\lambda_{\mathrm{th}} = h/\sqrt{2\pi m k_B T}$.

Notation for point defects. In this sheet, X^q denotes a defect X in the charge state q, E^f and G^f its formation energy and formation Gibbs energy, and $c = n/N$ its concentration per lattice site. The internal degeneracy of a defect is g_X, written g in [F], so that g keeps its meaning of the Gibbs free energy per particle. The Fermi level E_F is the electron chemical potential, measured from the valence-band maximum (VBM). The energy $e^2/(4\pi\epsilon_0) = 14.40$ eV Å is used for Coulomb estimates.

Exercise 1: The formation energy as a grand potential. A supercell of a crystal contains one defect X^q. The crystal exchanges atoms of species i with reservoirs at the chemical potentials μ_i and exchanges electrons with a reservoir at the chemical potential E_F.

(a) Which variables do the reservoirs fix, and which extensive quantities of the crystal therefore fluctuate? Which Legendre transform of U in Section 5 of the note has these fixed variables as natural variables?

(b) To create the defect, n_i atoms of species i enter the supercell ($n_i < 0$ if they leave). By how much does the free energy of the reservoirs change? Why is the right energy per atom μ_i and not the energy of an isolated atom?

(c) A charge state $q > 0$ means that q electrons have left the crystal for the electron reservoir. What energy does the reservoir gain? Combine parts (b) and (c) at $T = 0$ to obtain Eq. (1) of [F] without E_{corr}:

$$E^f[X^q] = E_{\mathrm{tot}}[X^q] - E_{\mathrm{tot}}[\mathrm{bulk}] - \sum_i n_i\mu_i + qE_F.$$

(d) Why is E^f a straight line in E_F, and what sets its slope? For a defect with several charge states, which charge state does the crystal choose at a given E_F? Compare this rule with the lower envelope $\mu = \min_\alpha \mu_\alpha$ of Section 10.3 of the note.

(e) Two defects A and B can bind into a complex AB. Show that the binding energy $E_b = E^f[A] + E^f[B] - E^f[AB]$, Eq. (2) of [F], does not depend on any μ_i. Why can a positive E_b still give few complexes in equilibrium? (Hint: compare the number of ways to place one complex with the number of ways to place two separate defects.)

Answer. (a) T, P, every μ_i and E_F are fixed, and the atom numbers and the electron number fluctuate. The potential is Φ generalized to several species and to electrons. (b) The reservoirs lose $\sum_i n_i\mu_i$, the free energy per atom of the reservoir phase. (c) The reservoir gains qE_F. (d) The slope is q, and the stable charge state has the lowest E^f, as in Fig. 1 of [F]. (e) Every n_i of AB equals the sum of those of A and B. Two separate defects have about N^2 placements, but a complex has only about zN, where z is the number of neighbor sites.
Related: Sections 4.2, 5.2, 5.5 and 10.3 of the note. *Source:* [F], Secs. I.B.1–2, Eqs. (1)–(2).

Exercise 2: Charge-state transition levels and the U parameter.

(a) At which Fermi level do the charge states q_1 and q_2 have equal formation energy? Derive Eq. (3) of [F]:

$$\varepsilon(q_1/q_2) = \frac{E^f(X^{q_1}; E_F = 0) - E^f(X^{q_2}; E_F = 0)}{q_2 - q_1}.$$

(b) For three charge states $+1$, 0 and -1, write $\varepsilon(+/0)$ and $\varepsilon(0/-)$ in terms of $E^f_q \equiv E^f(X^q; E_F = 0)$. Show that $U = E^f_{+1} + E^f_{-1} - 2E^f_0$, Eq. (43) of [F], equals $\varepsilon(0/-) - \varepsilon(+/0)$.

(c) If $U < 0$, is the neutral state ever the lowest line? At which Fermi level does the defect then switch directly from $+1$ to -1?

(d) In Section 10.3 of the note, a phase whose curve never touches the lower envelope is metastable everywhere. What plays the role of the temperature, and what the role of the phases, in the defect diagram?

(e) Why must $E^f(X^q)$ be computed with the atomic structure relaxed separately in each charge state for $\varepsilon(q_1/q_2)$ to be a thermodynamic level? (Exercise 11 contrasts this with optical levels.)

Answer. (a) Set $E^f(X^{q_1}) + q_1 E_F = E^f(X^{q_2}) + q_2 E_F$. (b) $\varepsilon(+/0) = E_0^f - E_{+1}^f$ and $\varepsilon(0/-) = E_{-1}^f - E_0^f$. (c) No. The switch occurs at $\varepsilon(+/-) = [\varepsilon(+/0) + \varepsilon(0/-)]/2$. (d) E_F plays the role of T, and the charge states play the role of the phases. A transition level is a "melting point" of the charge state. (e) A thermodynamic level compares two equilibrium states, each at the minimum of its own energy surface.
Related: Sections 10 and 10.3 of the note. *Source:* [F], Secs. I.B.3 and II.D, Eqs. (3), (43).

Exercise 3: Configurational entropy and the equilibrium concentration. A crystal has N sites, and n identical defects occupy them. Each defect has g_X internal orientations.

(a) Why does the number of arrangements equal $W = (g_X N)!/[(g_X N - n)!\, n!]$? Which factorial removes the distinction between identical defects, and which approximation gives $W \approx (g_X N)^n/n!$ for $n \ll N$?

(b) Apply Stirling's formula to $S^{\mathrm{conf}} = k_B \ln W$ and show that, per site, $S^{\mathrm{conf}}(c) = k_B[c - c\ln c + c\ln g_X]$, Eq. (7) of [F].

(c) Compare this entropy with the ideal mixing entropy $-k_B[c\ln c + (1-c)\ln(1-c)]$ of Section 11.7 of the note. Which term does the dilute limit drop?

(d) The Gibbs energy per site is $G(c) = G[\mathrm{bulk}] + cG^f - TS^{\mathrm{conf}}(c)$, Eq. (39) of [F]. What does $\partial G/\partial c = 0$ give? Derive $c^{\mathrm{eq}} = g_X \exp(-G^f/k_B T)$, Eq. (41).

(e) Show that G at equilibrium is $G[\mathrm{bulk}] - k_B T c^{\mathrm{eq}}$, Eq. (40). Why does the crystal always gain free energy by forming some defects, however large G^f is? Look at the slope of S^{conf} at $c \to 0$.

(f) Define the defect chemical potential $\mu_X = \partial G/\partial c$. Why is $\mu_X = 0$ at equilibrium? Which conservation law that holds for atoms fails for vacancies, and which other "particles" of Exercise 6 share this property?

Answer. (b) $\ln W \approx n \ln(g_X N) - n \ln n + n$. (c) The expansion $-(1-c)\ln(1-c) \approx c$ gives the term c, and only terms of order c^2 are dropped. (d) $G^f + k_B T \ln(c/g_X) = 0$. (e) $dS^{\text{conf}}/dc = -k_B \ln(c/g_X) \to +\infty$ as $c \to 0$. (f) Vacancies form by moving atoms to the surface, so their number is not conserved. Phonons, with $\mu = 0$, behave in the same way.
Related: Sections 2, 4.5, 5.6 and 11.7 of the note. *Source:* [F], Secs. II.A.1 and II.B.3, Eqs. (4)–(8), (39)–(41).

Exercise 4: Arrhenius plots and their extrapolation. Table I of [F] lists, for the vacancy in Al, $H^f = 0.76$ eV and $S^f = 2.4\,k_B$ from experiment at the melting temperature $T^m = 933$ K, and DFT values at 0 K and at T^m. The DFT values at T^m are not a single calculation at 933 K. They come from DFT free energies $G^f(T)$ with phonons and anharmonicity, fitted in the high-temperature range by a straight line exactly as the experimental data. Figure 1 shows the idea with a model $G^f(T)$ that reproduces the GGA values of Table I.

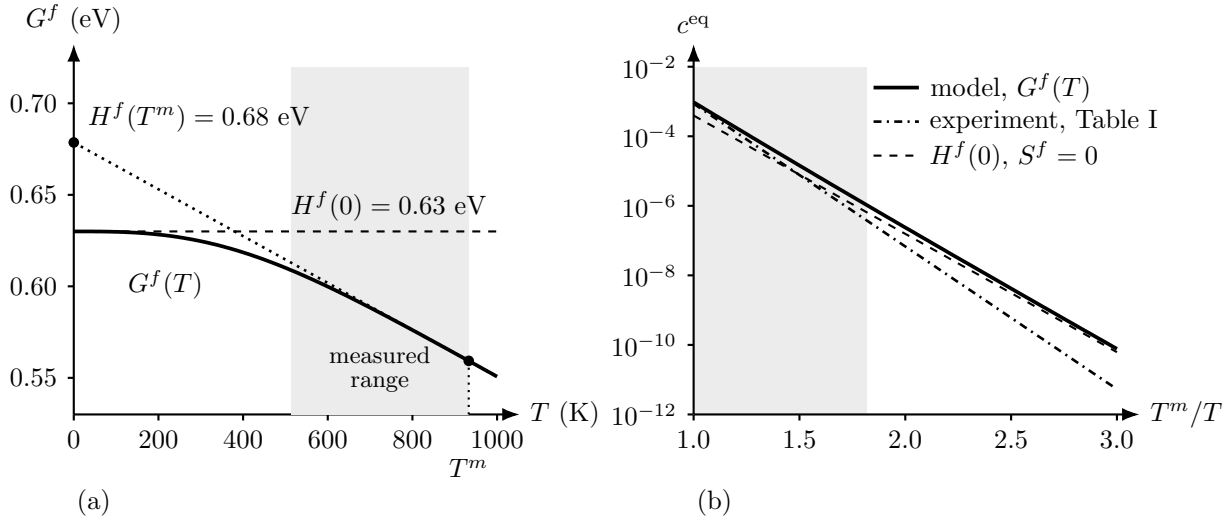


Figure 1: Vacancy formation in Al from a model that reproduces the GGA values of Table I of [F]. (a) Gibbs energy of formation $G^f(T)$ (solid), the 0 K enthalpy (dashed) and the tangent at T^m extrapolated to $T = 0$ (dotted). (b) Equilibrium concentration against T^m/T for the model, for the experimental straight line of Table I and for $H^f(0)$ with $S^f = 0$. Gray: temperatures above $0.55\,T^m$, the range of the experiments.

(a) If H^f and S^f do not depend on T, what does a plot of $\ln c^{\text{eq}}$ against $1/T$ look like? Which feature of the plot gives H^f, and which gives S^f? This is Eq. (42) of [F]. Which line in Fig. 1(b) has this form, and why does the model curve bend toward the dashed line at low temperature?

(b) Evaluate $c^{\text{eq}}(T^m)$ with $g_X = 1$. By what factor would c^{eq} be smaller if the vibrational

formation entropy were zero?

(c) Use $\partial^2 G^f/\partial T^2 = -C_P^f/T$, with the formation heat capacity C_P^f, as in Section 4 of the note. If $C_P^f > 0$, is $G^f(T)$ concave or convex?

(d) An experiment measures G^f only in the gray range of Fig. 1(a) and extrapolates the tangent line, the dotted line, to $T = 0$. Show that the intercept equals $H^f(T^m)$ and that the slope equals $-S^f(T^m)$. For a concave $G^f(T)$, which is larger, $H^f(T^m)$ or $H^f(0)$? Check the answer against the figure and against Table I of [F].

(e) In Fig. 1(b), the model and the straight experimental line nearly agree in the gray range but differ by about an order of magnitude at $T^m/T = 3$. What does this difference say about predicting defect concentrations at room temperature from high-temperature data?

Answer. (a) A straight line with the slope $-H^f/k_B$ and the intercept $\ln g_X + S^f/k_B$, the dash-dotted line. The model entropy falls to zero as $T \to 0$, so the model approaches the line with $H^f(0)$ and $S^f = 0$. (b) $H^f/k_B T^m = 9.45$ and $c^{\text{eq}} = 8.7 \times 10^{-4}$. Without S^f, it would be $e^{2.4} = 11$ times smaller. (c) Concave. (d) The tangent of a concave function lies above it, so $H^f(T^m) \geq H^f(0)$. Table I gives $0.68 > 0.63$ eV (GGA) and $0.78 > 0.68$ eV (LDA), both with anharmonicity. (e) An Arrhenius line fixed in the measured range does not predict the low-temperature concentration, because S^f and H^f change with T. The 0 K values with the temperature dependence of G^f are needed.
Related: Sections 3.3, 4.5 and 5.3 of the note. *Source:* [F], Secs. II.B.3 and II.C.1, Eq. (42), Table I.

Exercise 5: Electronic entropy and the Fermi–Dirac distribution. Electrons are fermions, so each single-particle orbital holds either 0 or 1 electron of a given spin. Non-interacting orbitals are independent.

(a) Treat one orbital of energy ϵ_i as a grand canonical system with $N_i \in \{0, 1\}$, in contact with an electron reservoir at $\mu = E_F$. What is Ξ_i? Show that the mean occupation is the Fermi–Dirac function $f_i = 1/[e^{\beta(\epsilon_i - E_F)} + 1]$.

(b) What is the Gibbs–Shannon entropy of a two-outcome distribution with the probabilities f_i and $1 - f_i$? Why does the total entropy add over orbitals? Derive Eq. (11) of [F]:

$$S^{\text{el}} = -k_B \sum_i \left[(1 - f_i) \ln(1 - f_i) + f_i \ln f_i \right].$$

(c) Compare part (b) with the ideal mixing entropy of Section 11.7 of the note. What is "mixed" here?

(d) Near E_F, let the density of states be a constant D_F, counting both spins. Turn the sum into an integral with $x = \beta(\epsilon - E_F)$ and use

$$\int_{-\infty}^{\infty} dx\,[-f\ln f - (1-f)\ln(1-f)] = \frac{\pi^2}{3}, \qquad f = \frac{1}{e^x+1},$$

to show that $S^{\rm el} = (\pi^2/3)k_B^2 T D_F$.

(e) The excitation energy is $\Delta E^{\rm el} = (\pi^2/6)(k_BT)^2 D_F$. Show that $\tilde{F}^{\rm el} = \Delta E^{\rm el} - TS^{\rm el} = -\frac{1}{2}TS^{\rm el}$, Eq. (13) of [F]. Why is this contribution small in a semiconductor but not in a metal?

(f) A neutral defect has one unpaired electron in a localized gap orbital that holds either spin. Which part of $S^{\rm el}$ survives as $T \to 0$, and where does it reappear in Exercise 3?

Answer. (a) $\Xi_i = 1 + e^{-\beta(\epsilon_i - E_F)}$ and $f_i = -\partial \ln \Xi_i / \partial(\beta\epsilon_i)$. (b) $-k_B[f\ln f + (1-f)\ln(1-f)]$, and independent systems have additive entropies (Section 3.2 of the note). (c) Occupied and empty orbitals are "mixed". (d) The integral is $\pi^2/3 = 3.290$. (e) $D_F \approx 0$ in the band gap of a semiconductor. (f) $k_B \ln 2$, which is the spin degeneracy $g_X = 2$ in $S^{\rm conf}$.
Related: Sections 1.1, 3.2, 5 and 11.7 of the note. *Source:* [F], Sec. II.A.2, Eqs. (10)–(13).

Exercise 6: Vibrational free energy and the Bose–Einstein distribution. Near equilibrium, the energy of the ions is quadratic in the displacements $u_k = R_k - R_k^0$, Eq. (9) of [F].

(a) Newton's equations read $M_k \ddot{u}_k = -\sum_l \Phi_{kl} u_l$, with the force constants $\Phi_{kl} = \partial^2 F^{\rm el}/\partial R_k \partial R_l$. Substitute $u_k = w_k e^{-i\omega t}/\sqrt{M_k}$. Why does the dynamical matrix $D_{kl} = \Phi_{kl}/\sqrt{M_k M_l}$, Eq. (14) of [F], appear, and why are its eigenvalues ω_i^2?

(b) A quantum oscillator has the levels $(n + \frac{1}{2})\hbar\omega$, $n = 0, 1, 2, \ldots$. Sum the geometric series for Z. Show that $F = \hbar\omega/2 + k_BT\ln(1 - e^{-\beta\hbar\omega})$, one term of Eq. (17) of [F].

(c) Show that the mean energy is $(\frac{1}{2} + \bar{n})\hbar\omega$ with $\bar{n} = 1/(e^{\beta\hbar\omega} - 1)$, Eq. (16) of [F]. Read $\bar{n}$ as the mean number of phonons in the mode. Compare $\bar{n}$ with f of Exercise 5. Which sign changed, and why is the phonon chemical potential zero?

(d) Find the high-temperature limit of F. Show that the vibrational formation entropy becomes $S^f_{\rm vib} = k_B \sum_i \ln(\omega_i^{\rm bulk}/\omega_i^{\rm def})$, with equal numbers of modes on both sides.

(e) Around a vacancy, the bonds of the neighbor atoms are broken. Do their modes soften or stiffen, and what is the sign of $S^f_{\rm vib}$? Estimate $S^f_{\rm vib}$ if the 36 modes of the 12 nearest neighbors in an fcc metal soften by 5%, and compare with $S^f = 2.4\,k_B$ for Al.

Answer. (a) The substitution turns the equations into the eigenvalue problem $Dw = \omega^2 w$, Eq. (15) of [F]. D is real and symmetric, and stability makes it positive definite. (b) $Z = e^{-\beta\hbar\omega/2}/(1-e^{-\beta\hbar\omega})$. (c) -1 replaces $+1$ in the denominator, because any number of phonons can occupy a mode. Phonon numbers are not conserved. (d) $F \to k_B T \ln(\hbar\omega/k_B T)$. (e) The modes soften, and $S^f_{\rm vib} > 0$. The estimate gives $36 \ln(1/0.95) = 1.8\, k_B$.
Related: Sections 3, 3.3 and 4.5 of the note. *Source:* [F], Sec. II.A.3, Eqs. (9), (14)–(17).

Exercise 7: Formation volume and pressure. At finite T and P, the formation Gibbs energy of Eq. (23) of [F] contains the term PV^f, with the formation volume $V^f = \Omega' - \Omega$.

(a) Which derivative of $G^f(T,P)$ gives V^f? Use Section 5.3 of the note.

(b) Show that $(\partial \ln c^{\rm eq}/\partial P)_T = -V^f/k_B T$. For a vacancy in Al with $V^f = 11$ Å^3, by what factor does 1 GPa change $c^{\rm eq}$ at 900 K?

(c) A supercell of fixed volume V_0 prevents the relaxation V^f. Use $B = -V\,\partial P/\partial V$ to show that the elastic energy stored is $\Delta E \approx B(V^f)^2/(2V_0)$, Eq. (57) of [F]. Evaluate ΔE for $B = 100$ GPa, $V^f = 10$ Å^3 and $V_0 = 1000$ Å^3.

(d) How does ΔE scale with the supercell size L? Why does stability, $B > 0$, make the constant-volume energy higher than the relaxed energy?

Answer. (a) $V^f = (\partial G^f/\partial P)_T$. (b) $\Delta \ln c = -0.89$, so c drops by a factor of 2.4. (c) $\Delta E = \int_0^{V^f} B\,(V/V_0)\,dV = 31$ meV. (d) $\Delta E \propto 1/L^3$. The constrained state lies on a convex $F(V)$ away from the minimum.
Related: Sections 4.1, 4.6 and 5.3 of the note. *Source:* [F], Secs. II.B.1 and III.D, Eqs. (23), (57).

Exercise 8: Chemical potentials and the stability window. A nitrogen vacancy $V_{\rm N}$ forms in GaN, which coexists either with Ga metal or with N_2 gas.

(a) In GaN at equilibrium, why must $\mu_{\rm Ga} + \mu_{\rm N} = \mu({\rm GaN})$, Eq. (26) of [F]? Use the Euler relation $G = \sum_i \mu_i N_i$ of Section 5.5 of the note.

(b) If $\mu_{\rm Ga}$ exceeded $\mu_{\rm Ga}$(Ga metal), what would form? Derive the bounds of Eqs. (27)–(30) of [F] and show that

$$\Delta_f G^0({\rm GaN}) \le \Delta\mu_{\rm Ga} \le 0, \qquad \Delta\mu_{\rm N} = \Delta_f G^0({\rm GaN}) - \Delta\mu_{\rm Ga},$$

in the notation of Eqs. (31)–(34).

(c) Draw $g(X)$ for the Ga–N system at fixed (T, P), with three line phases at $X = 0$ (Ga), $X = 1/2$ (GaN) and $X = 1$ (N_2), and construct the convex hull of Section 11.4 of the note. Which tangent lines touch the GaN point? Show that their intercepts at $X = 0$ and $X = 1$ sweep exactly the window of part (b).

(d) The formation of V_N removes one N atom, $n_N = -1$. How much larger is $E^f(V_N)$ under N-rich than under Ga-rich conditions?

(e) For N_2 gas, the molecule has the ground energy $E(N_2)$ and the rotational partition function $z_{rot} = k_BT/(\sigma B_0)$ for $k_BT \gg B_0$. Start from $\mu = k_BT \ln(P\lambda_{th}^3/k_BT)$ of Exercise 13 of `exercise.pdf` and multiply Z_N by z_{int}^N. Derive Eq. (35) of [F]:

$$2\mu_N = E(N_2) + k_BT \ln \frac{PV_Q}{k_BT} + k_BT \ln \frac{\sigma B_0}{k_BT} + \mu_{vib}, \qquad V_Q = \lambda_{th}^3.$$

(f) Raising $P(N_2)$ at fixed T moves μ_N in which direction? How does $c(V_N)$ respond?

Answer. (b) Excess Ga would precipitate as metal, and excess N would leave as N_2. (c) The tangents through the GaN point rotate between the slopes of the Ga–GaN and GaN–N_2 segments of the hull. (d) By $|\Delta_f G^0(\text{GaN})|$, about 1 eV. (e) $\mu = -k_BT \ln z_{int} + k_BT \ln(P\lambda_{th}^3/k_BT)$, with $z_{int} = e^{-\beta E(N_2)} z_{rot} z_{vib}$. (f) μ_N rises by $\frac{1}{2} k_BT\, \Delta \ln P$, and $c(V_N) \propto P^{-1/2}$.
Related: Sections 5.5, 5.7, 11.3 and 11.4 of the note. *Source:* [F], Sec. II.B.2, Eqs. (26)–(35).

Exercise 9: Charge neutrality and the Fermi level. A semiconductor has the band gap E_g, parabolic bands with the effective masses m_e and m_h, and dilute charged defects. Electrons and holes follow Exercise 5.

(a) Why must the crystal as a whole be neutral, Eq. (36) of [F]? Which unknown does this condition fix, once all $c(X^q)$ are written as functions of E_F?

(b) The conduction band has the density of states per volume

$$D_c(E) = \frac{1}{2\pi^2} \left(\frac{2m_e}{\hbar^2} \right)^{3/2} \sqrt{E - E_c},$$

with both spins included. When $E_c - E_F \gg k_BT$, which term of the Fermi–Dirac denominator can be dropped? Show that $n_e = N_c e^{-(E_c - E_F)/k_BT}$ with $N_c = 2(m_e k_BT/2\pi\hbar^2)^{3/2}$. Compare with Eq. (38) of [F], which counts one spin.

(c) Evaluate N_c for m_e equal to the free-electron mass at 300 K.

(d) Without defects, $n_e = n_h$. Show that $E_F = E_g/2 + \frac{3}{4} k_BT \ln(m_h/m_e)$.

(e) A donor D^+ has $E^f = E_D + E_F$, and a native acceptor A^- has $E^f = E_A - E_F$, with $c \propto e^{-E^f/k_BT}$. If the defects outnumber the free carriers, where does neutrality pin E_F? Locate this point in a plot of E^f against E_F.

(f) Raising the dopant chemical potential μ_D lowers E_D by the same amount. Show that E_F then rises only by half of the increase of μ_D. Why does this effect, self-compensation, limit doping (Sec. I.A.2 of [F])?

Answer. (a) The crystal has no external source of charge, so neutrality fixes E_F. (b) $f \approx e^{-\beta(E-E_F)}$, and the integral is a Gaussian integral. (c) $N_c = 2.5 \times 10^{19}$ cm^{-3}. (d) $N_c/N_v = (m_e/m_h)^{3/2}$. (e) $c(D^+) = c(A^-)$ gives $E_F = (E_A - E_D)/2$, where the two lines cross. (f) $E_F = (E_A - E_D)/2$ rises by $\Delta\mu_D/2$. Each added donor is compensated by a new acceptor, so the carrier density barely grows. *Related:* Sections 5, 5.5 and 10.3 of the note. *Source:* [F], Secs. I.A.2 and II.B.2c, Eqs. (36)–(38).

Exercise 10: Vacancies close to the melting point. Section V.B of [F] asks whether vacancies cause the rise of the heat capacity of Al close to T^m.

(a) From Eq. (40) of [F], the vacancy contribution to the Gibbs energy per site is $\Delta G = -k_BTc^{\rm eq}(T)$. Use $H = -T^2\,\partial(G/T)/\partial T$ to show that $\Delta H = c^{\rm eq}H^f$. Why is this result expected?

(b) Show that $\Delta C_P = k_Bc^{\rm eq}(H^f/k_BT)^2$.

(c) The number of vacancies in N sites is a sum of nearly independent site occupations. What is its variance? Use $\mathrm{Var}(U) = k_BT^2C$ of Section 3.3 of the note to obtain part (b) a second way.

(d) Evaluate ΔC_P per atom for Al at T^m with the data of Exercise 4, and compare it with $3k_B$.

(e) Which other response function of Section V.B of [F] should reveal vacancies, and why does it depend on V^f?

Answer. (a) Each vacancy carries the enthalpy H^f. (c) $\mathrm{Var}(n) \approx n$, so $\mathrm{Var}(U) = (H^f)^2n$. (d) $\Delta C_P = 0.078\,k_B$ per atom, about 2.6% of $3k_B$. (e) The thermal expansion coefficient, because each vacancy adds V^f to the volume.
Related: Sections 3.3, 4.6 and 5.3 of the note. *Source:* [F], Secs. II.B.3 and V.B, Eq. (40).

Exercise 11: Optical levels and the configuration-coordinate diagram. In Fig. 8 of [F], the substitutional acceptor N_O in ZnO has the absorption energy $E_{\rm abs} = 2.4$ eV and the emission energy $E_e = 1.7$ eV. ZnO has $E_g = 3.4$ eV.

(a) Model the two states N_O^- and $\mathrm{N}_\mathrm{O}^0 + e_\mathrm{CB}^-$ by parabolas $\frac{1}{2}K(Q-Q_-)^2$ and $E_\mathrm{ZPL} + \frac{1}{2}K(Q-Q_0)^2$ with equal curvature K. Why do absorption and emission occur at fixed Q (Franck–Condon principle)?

(b) Show that $E_\mathrm{abs} = E_\mathrm{ZPL} + E_\mathrm{rel}$ and $E_e = E_\mathrm{ZPL} - E_\mathrm{rel}$, with $E_\mathrm{rel} = \frac{1}{2}K(Q_- - Q_0)^2$. Find E_rel, the Stokes shift and E_ZPL.

(c) The zero-phonon line equals $E_g - \varepsilon(0/-)$. Find $\varepsilon(0/-)$ and compare with 1.3 eV in [F].

(d) Why does a Fermi-level-dependent concentration use the thermodynamic level, whereas an absorption spectrum uses the optical level? Compare the time scales of electronic and ionic motion.

(e) For a typical phonon energy $\hbar\omega = 40$ meV, how many phonons does the relaxation emit? Why does this number broaden the emission band?

Answer. (b) $E_\mathrm{rel} = 0.35$ eV, the Stokes shift is 0.7 eV, and $E_\mathrm{ZPL} = 2.05$ eV. (c) $\varepsilon(0/-) = 1.35$ eV. (d) Electronic transitions take about 10^{-15} s, ionic relaxation about 10^{-13} s. (e) About $E_\mathrm{rel}/\hbar\omega \approx 9$ phonons.
Related: Sections 3 and 10 of the note. *Source:* [F], Sec. II.E, Fig. 8, Eq. (43).

Exercise 12: Diffusion by vacancy hopping. Section I.A.3 of [F] states that impurities usually diffuse with the help of vacancies or interstitials. This exercise builds the diffusion coefficient from the canonical ensemble.

(a) An atom makes uncorrelated jumps of length a in random directions, Γ jumps per unit time. Why do the cross terms $\langle \vec{a}_j \cdot \vec{a}_k \rangle$, $j \neq k$, vanish? Show that $\langle r^2 \rangle = \Gamma a^2 t$, and define D by $\langle r^2 \rangle = 6Dt$.

(b) Check that $c(\vec{r},t) \propto t^{-3/2} e^{-r^2/4Dt}$ solves $\partial c/\partial t = D\nabla^2 c$ and has $\langle r^2 \rangle = 6Dt$.

(c) In one dimension, an atom sits in a well of curvature K and mass m, so that the attempt frequency is $\nu = \sqrt{K/m}/2\pi$, and a barrier of height E_m separates the well from the next one. In canonical equilibrium, what is the probability density of finding the atom at the barrier top, normalized over the well? What is the mean forward velocity $\langle v\,\theta(v) \rangle$ from the Maxwell distribution? Multiply the two to show that the jump rate is $\Gamma_0 = \nu e^{-E_m/k_B T}$, the result of transition-state theory.

(d) An atom in a crystal jumps only into a neighboring vacancy. With z neighbors and the vacancy concentration c_v^eq of Exercise 3, show that the self-diffusion coefficient is

$$D = \frac{f_c\, z\, a^2 \nu}{6} \exp\!\left(\frac{S^f}{k_B}\right) \exp\!\left(-\frac{H^f + E_m}{k_B T}\right),$$

where the correlation factor f_c (0.78 for fcc) corrects for jumps straight back into the same vacancy.

(e) Which energy does an Arrhenius plot of D measure? For Al, with $H^f = 0.67$ eV and $E_m = 0.61$ eV, compare with the measured activation energy of about 1.3 eV.

(f) Section I.A.3 of [F] says that interstitials move more readily than vacancies but are less abundant. Which sum decides the mechanism that dominates diffusion?

(g) In a non-uniform crystal, the flux is $J = -(Dc/k_BT)\,\partial\mu/\partial x$. For a dilute species with $\mu = k_BT \ln c + \text{const}$, recover Fick's law. Why does this result agree with the statement of Section 11.11 of the note that no flux flows through a two-phase zone?

Answer. (a) Independent jumps have zero mean. (c) The density at the barrier top, normalized over the well, is $\sqrt{K/2\pi k_BT}\, e^{-\beta E_m}$, and $\langle v\,\theta(v)\rangle = \sqrt{k_BT/2\pi m}$. (e) $H^f + E_m = 1.28$ eV. (f) $H^f + E_m$ for each mechanism. (g) $J = -D\,\partial c/\partial x$. Inside a two-phase zone μ is uniform, so $J = 0$. *Related:* Sections 3, 11.4 and 11.11 of the note. *Source:* [F], Secs. I.A.3 and II.B.3, Eq. (41).

Exercise 13: Charged defects in a periodic supercell. A supercell calculation repeats one charged defect q on a cubic lattice of period L in a medium of dielectric constant ε.

(a) Why does the electrostatic energy of a periodic array of net charges diverge? Why does a uniform compensating background remove the divergence? What happens to the $\vec{G} = 0$ Fourier component of the potential?

(b) On dimensional grounds, how must the spurious interaction scale with q, ε and L? For a simple cubic array of point charges in a background, the Madelung energy is $E_M = \alpha_M q^2/(2\varepsilon L)$ with $\alpha_M = 2.8373$. Evaluate E_M for $q = 1$, $\varepsilon = 10$ and $L = 12$ Å.

(c) The computational cost grows as the cube of the number of atoms. By what factor does the cost grow if the error is halved? This is the estimate of Sec. I.C.2 of [F].

(d) The energy behaves as $E(L) = E_\infty + a_1/L + a_3/L^3$, with an L^{-3} term from the quadrupole moment. How many supercell sizes determine E_∞? How many if a_1 is taken from the Madelung energy?

(e) Why is the Fermi level of Exercise 1 undefined in a charged supercell until the average electrostatic potential is aligned with that of the bulk (Fig. 12 of [F])?

Answer. (a) The Coulomb sum over the net charges grows with the number of cells. The background sets the average charge, the $\vec{G} = 0$ term, to zero. (b) $E_M \propto q^2/(\varepsilon L)$, and $E_M = 0.17$ eV, larger than

many target accuracies. (c) Halving the error doubles L, so the number of atoms grows 8 times and the cost $8^3 = 512$ times. (d) Three sizes, or two with a_1 known. (e) E^f contains qE_F, and E_F is measured from the VBM, whose absolute energy depends on the reference of the potential. *Related:* Section 5.2 of the note. *Source:* [F], Secs. I.C.2 and III.C.